Real-Time Multi-Vehicle Detection and Tracking Using YOLOv8 and DeepSORT for Autonomous Systems

Real-Time → The system operates instantly without significant delay.

Multi-Vehicle → It detects and tracks multiple vehicles simultaneously.

Detection and Tracking → The system identifies vehicles and follows them as they move.

Using YOLOv8 and DeepSORT → YOLOv8 is the deep learning model for object detection, while DeepSORT is the tracking algorithm.

For Autonomous Systems → The system is designed for self-driving cars, smart traffic monitoring, and similar applications.

Abstract :-

This project presents a real-time vehicle detection and tracking system using advanced deep learning techniques. The system integrates YOLOv8, a state-of-the-art object detection model, with DeepSORT, a robust tracking algorithm, to achieve precise multi-vehicle tracking in video streams. The primary objective is to identify and track multiple vehicles in real time, ensuring continuous monitoring across video frames.

YOLOv8 enables high-speed and accurate vehicle detection, effectively identifying vehicles such as cars, buses, and trucks with high confidence. DeepSORT assigns unique IDs to detected vehicles, ensuring consistent tracking even in dense traffic conditions. This implementation is optimized for computational efficiency while maintaining high accuracy. The model processes video frames at a high frame rate, ensuring smooth real-time tracking. The system also provides insights into vehicle movement, which can be beneficial for traffic analysis, congestion management, and automated surveillance applications.

The developed framework can be integrated into urban traffic control systems, smart city solutions, and autonomous vehicle navigation. The project enhances road safety, traffic monitoring, and transportation management, making it a valuable tool for intelligent mobility solutions.

Keywords :-

Real-time vehicle tracking, YOLOv8, DeepSORT, object detection, intelligent transportation system, autonomous vehicles, deep learning, multi-object tracking, traffic monitoring, smart surveillance.